

KITABU QUEST

S6

Subsidiary Mathematics LFK HLP HGL

Mascot: mountain gorilla

Scene: students using applied mathematics for humanities and language combinations with a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S6 Subsidiary Mathematics LFK HLP HGL

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S6 learners study Subsidiary Mathematics LFK HLP HGL one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Bi-variate statistics
2. Counting Techniques
3. Elementary probability

As you finish each topic, return to the success criteria and correct one answer before moving on.

Bi-variate statistics: Lesson Opener

Learning focus: Bi-variate statistics

Country link: a Rwanda school, market, farming, building, transport, or data problem for Bi-variate statistics.

By the end of this topic, you should be able to:

- extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines

Inquiry questions:

- How can bi-variate statistics help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Bi-variate statistics: Learn

Bi-variate statistics teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Bi-variate statistics
- Bi-variate
- statistics
- method
- model
- unit
- answer
- check

Bi-variate statistics: Worked Example

Worked example for Bi-variate statistics: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Bi-variate statistics: Vocabulary And Study Skill

Use these words and study moves before you practise Bi-variate statistics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Bi-variate statistics
- Bi-variate
- statistics
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Bi-variate statistics without a clear example, method, or evidence.

Bi-variate statistics: Guided Activity

Guided activity for Bi-variate statistics:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Bi-variate statistics: Practice And Correction

Practice for Bi-variate statistics:

Main outcome: extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to bi-variate statistics and solve it.

Bi-variate statistics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Bi-variate statistics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Bi-variate statistics: Outcome Check

Outcome check for Bi-variate statistics: Extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Counting Techniques: Lesson Opener

Learning focus: Counting Techniques

Country link: a Rwanda school, market, farming, building, transport, or data problem for Counting Techniques.

By the end of this topic, you should be able to:

- use counting techniques to determine the number of possible outcomes of events occurring in real life

Inquiry questions:

- How can counting techniques help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Counting Techniques: Learn

Counting Techniques teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Counting Techniques
- Counting
- Techniques
- method
- model
- unit
- answer
- check

Counting Techniques: Worked Example

Worked example for Counting Techniques: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Counting Techniques: Vocabulary And Study Skill

Use these words and study moves before you practise Counting Techniques. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Counting Techniques
- Counting
- Techniques
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Counting Techniques without a clear example, method, or evidence.

Counting Techniques: Guided Activity

Guided activity for Counting Techniques:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Counting Techniques: Practice And Correction

Practice for Counting Techniques:

Main outcome: use counting techniques to determine the number of possible outcomes of events occurring in real life.

1. Solve one counting techniques question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to counting techniques and solve it.

Counting Techniques: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Counting Techniques.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Counting Techniques: Outcome Check

Outcome check for Counting Techniques: Use counting techniques to determine the number of possible outcomes of events occurring in real life.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Lesson Opener

Learning focus: Elementary probability

Country link: a Rwanda school, market, farming, building, transport, or data problem for Elementary probability.

By the end of this topic, you should be able to:

- determine the probability of possible outcomes of events occurring under equally likely assumptions

Inquiry questions:

- How can elementary probability help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Learn

Elementary probability teaches how likely an event is.

Core idea:

- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:

- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Elementary probability: Worked Example

Worked example for Elementary probability: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Elementary probability: Vocabulary And Study Skill

Use these words and study moves before you practise Elementary probability. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Elementary probability without a clear example, method, or evidence.

Elementary probability: Guided Activity

Guided activity for Elementary probability:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Elementary probability: Practice And Correction

Practice for Elementary probability:

Main outcome: determine the probability of possible outcomes of events occurring under equally likely assumptions.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:

Answer check: certain means must happen; impossible means cannot happen.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to elementary probability and solve it.

Elementary probability: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Elementary probability.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Outcome Check

Outcome check for Elementary probability: Determine the probability of possible outcomes of events occurring under equally likely assumptions.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Bi-variate statistics: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Bi-variate statistics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Counting Techniques: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Counting Techniques that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Elementary probability: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Elementary probability that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Bi-variate statistics: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Bi-variate statistics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics LFK HLP HGL.

- Bi-variate statistics: Explain this term using an example from the book, your school, or your community.
- Counting Techniques: Explain this term using an example from the book, your school, or your community.
- Elementary probability: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics LFK HLP HGL answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics LFK HLP HGL that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.